iModMix, a network-based framework and R package for multi-omics analysis

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Introduction

- iModMix is an R package for analyzing multi-omic datasets.
- iModMix uses graphical lasso to build biologically meaningful networks without the “hairball problem”.
- Fuzzy nearest neighbor clustering is applied to topological overlap measures (TOMs) for identifying groups of related features (“modules”).
- This fuzzy approach allows features to belong to more than one module, as for example metabolites can belong to multiple pathways.
- Modules are empirically derived from data and are not constrained by pre-defined pathways (e.g., KEGG).
- Importantly, modules from multi-omics datasets can be related to a phenotype and also between omic data types using module eigenfeatures.
- We applied iModMix to a novel proteometabolomics dataset of KRAS mutant lung tumors from genetically altered mice (n = 20).

Graphical Methods

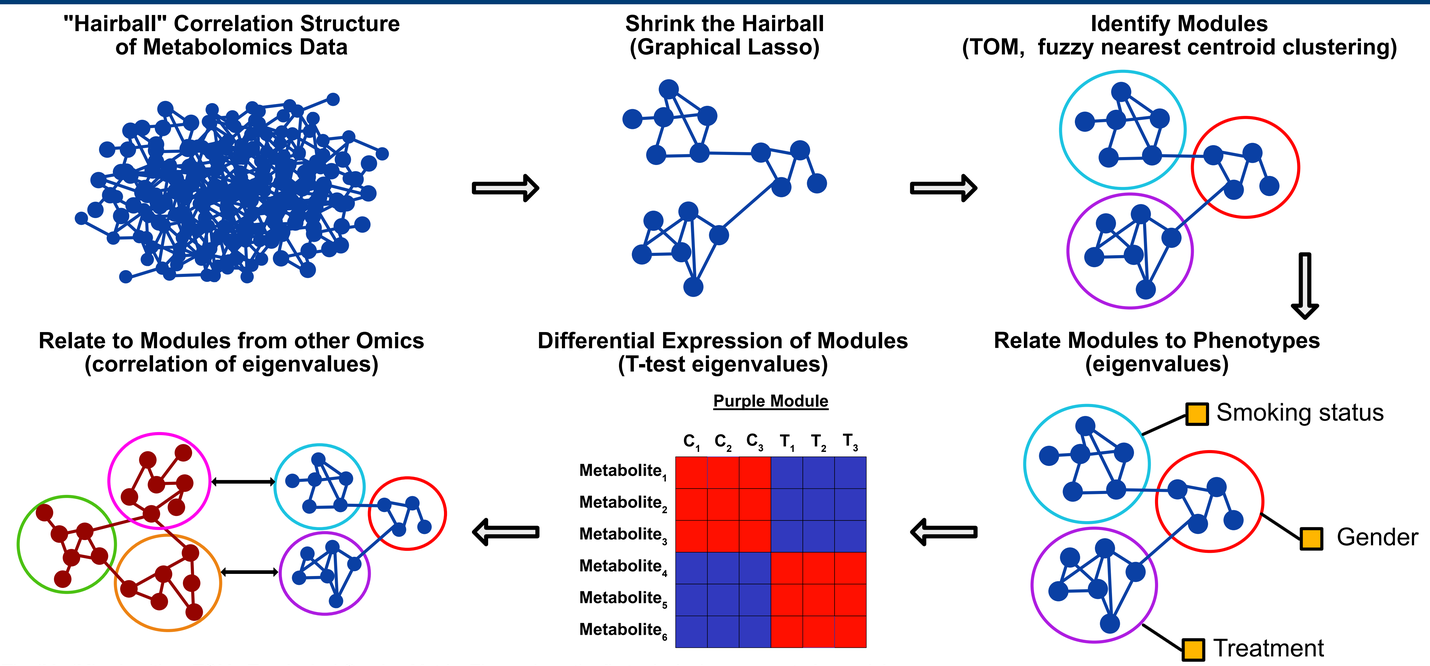

Figure 1 – The iModMix algorithm. TOM - Topological Overlap Matrix; Eigenvalue - the first principal component of a module.

Results – Module Identification

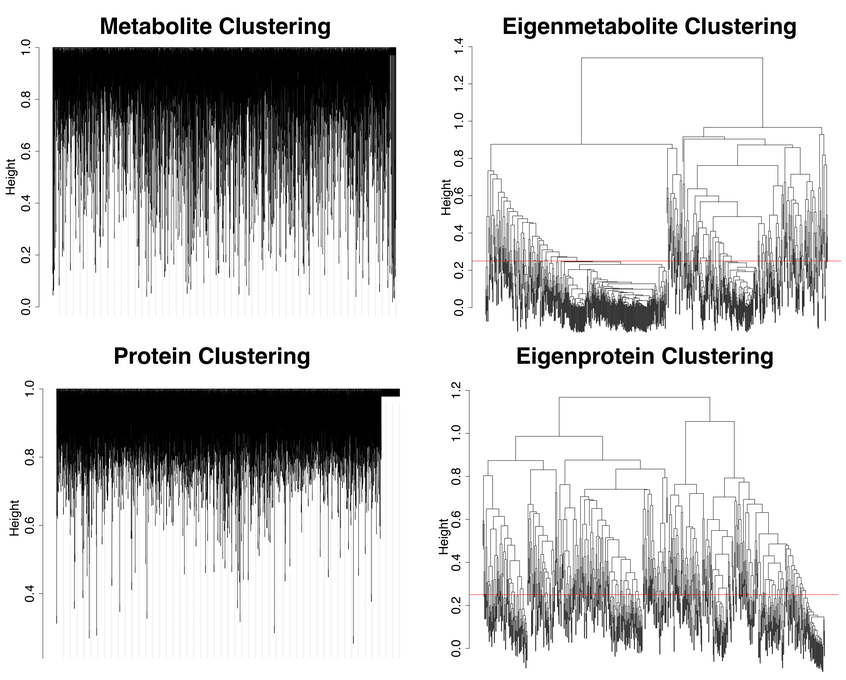

Figure 2 – (Left) Clustering dendrograms of metabolites and proteins using TOM-based dissimilarity. (Right) Eigenmetabolites/eigenproteins reduce dimensionality and can be used for testing differential expression of constituent metabolites/proteins.

Results – Differentially Expressed Modules

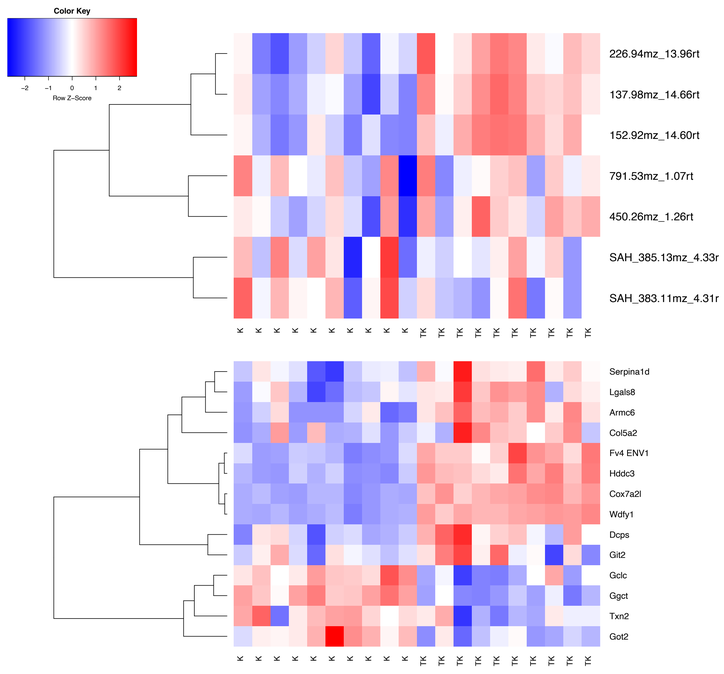

Figure 3 – Differentially expressed eigenmetabolites/eigenproteins relate back to differential expression of constituent proteins/metabolites in KRAS mutant tumors (K) compared to KRAS and TAp73 mutant tumors (TK).

Results – Integrative Module Analysis

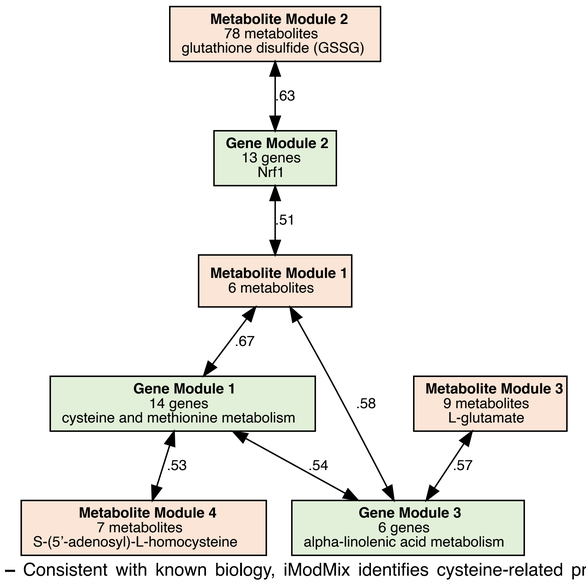

Figure 4 – Consistent with known biology, iModMix identifies cysteine-related protein and metabolite modules. Edge values are eigenfeature correlations between modules.

Conclusion

Ongoing work is investigating iModMix’s ability to identify unknown metabolites based on associations with known modules. iModMix’s data agnostic design means it can handle any combination of omics expression data including features with unknown identity/function.